

Phil Piwonka

Software Engineer

📍 Seattle, Washington, US ✉ jobs@pdp.dev 🌐 pdp.dev 🐙 wnka

I love computers and writing software for them.

WORK

Principal Software Development Engineer

Sep 2025 – Present

AWS Bedrock

Selected to join the small, senior engineering team building Project Mantle, a next-generation inference engine for Amazon Bedrock.

- Focused on the OpenAI Responses API, working with OpenAI to provide compatibility and behavioral parity with the first-party API and implementation.
- Implemented server-side tool calling for the Responses API using connectors to AWS Lambda and Amazon Bedrock AgentCore Gateway, enabling customers to bring existing MCP infrastructure to inference.
- Implemented Web Search for OpenAI frontier models in the Responses API, integrating an existing Amazon search service into AWS with IAM authorization, limit enforcement, fairness algorithms, and monitoring.
- Gave internal talks and helped teams across Amazon understand how to apply AI-driven development.
- Partnered with Builder Tools to improve build times for the Mantle monorepo, supporting rapid AI-driven development while working through the constraints of existing engineering tools.

Principal Software Development Engineer

Apr 2016 – Sep 2025

AWS Lambda

Worked across AWS Lambda infrastructure, focusing on utilization, efficiency, capacity planning, and major platform launches as the service grew rapidly.

- Designed and implemented a new Lambda Placement service and engine to manage oversubscription, heat management, and efficient workload placement across the fleet.
- Built simulators and utilization models to evaluate placement strategies and determine workload entitlements, informing capacity-planning decisions and system design.
- Evolved Placement through Lambda's multi-year migration to Firecracker, adapting capacity management and tenancy as the service grew rapidly and its virtualization model changed.
- Extended Placement to handle heterogeneous capacity and workload constraints, including Intel and AMD processors, AWS Graviton, NVMe- and EBS-backed instances, and increasing Lambda memory sizes from 1.5 GB to 10 GB.
- Launched AWS Lambda support for container images, expanding the platform to support container-based workloads and contributing to the award-winning research published in the On-demand Container Loading in AWS Lambda paper.
- Launched Lambda SnapStart to reduce cold-start times and improve startup performance for latency-sensitive applications.

- Served as an Operations bar raiser across Lambda, ECS Fargate, and App Runner; participated in on-call throughout, later joining escalation on-call because of broad system knowledge and operational-event expertise.

Senior Software Development Engineer

Sep 2010 – Apr 2016

Amazon Kindle

Selected to join the original Kindle Fire team; built cross-platform reader experiences, device UI, and backend services integrating Goodreads with Kindle devices.

- Helped develop ACX, an HTML5 framework used to deliver Kindle Reader features across Qt-based Mac and PC applications, Android, and iOS.
- Delivered reader experiences including Wikipedia Lookup, passage translation, X-Ray, and End Actions, enabling in-book discovery, contextual information, and post-book recommendations.
- Developed the original Kindle Fire homescreen launcher and carousel, using custom GPU-accelerated rendering for content art and drop shadows while maintaining responsive performance during rapid scrolling and navigation across books, music, and video content.
- Built GRoK (Goodreads on Kindle), the server-side integration that brought Goodreads experiences to first-party Kindle e-readers and Fire tablets by bringing the Goodreads Ruby on Rails stack into Amazon infrastructure and creating services to cache, front, and protect the Goodreads backend from Kindle traffic.
- Prepared for and participated in Christmas traffic gamedays and production on-call, including the GRoK launch as part of the Kindle device registration flow that linked Goodreads and Kindle accounts.
- Helped modernize Goodreads infrastructure by decomposing functionality from its Ruby on Rails monolith into services and using DynamoDB to reduce load on SQL databases.
- Launched Goodreads Deals, a personalized daily email campaign matching discounted ebooks with titles on customers' Want to Read shelves, with batch campaign generation decoupled from online customer traffic.

Software Development Engineer

Nov 2006 – Sep 2010

Microsoft

Member of the SQL Server team

- Developed database providers for the Microsoft Sync Framework, enabling peer-to-peer synchronization capabilities across applications, services, and devices.
- Delivered provider integration in Visual Studio 2008 and 2010 and designed new Merge Replication features for SQL Server 2008.
- Supported enterprise customers by diagnosing urgent issues, gathering feedback, and helping onboard new developers.

Software Engineer

Jun 2003 – Oct 2006

Lockheed Martin

Worked on near-realtime processing systems

- Designed and tested mission-critical numerical and clustering algorithms in a near-realtime, high-performance UNIX parallel computing environment.
 - Scaled algorithms with multithreading and shared-memory programming, and built C++, Matlab, and Perl tools for prototyping and validation.
 - Delivered work to multiple customer organizations and led several software development tasks.
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PUBLICATIONS

[Firecracker: Lightweight Virtualization for Serverless Applications](#)

USENIX Association · 2020

We describe how specializing for serverless informed the design of Firecracker, and what we learned from seamlessly migrating AWS Lambda customers to Firecracker.

[On-demand Container Loading in AWS Lambda](#)

USENIX Association · Jul 2023

Awarded Best Paper at USENIX ATC'23! - We describe the storage and caching system we built, optimized for delivering container images on-demand, and our experiences designing, building, and operating it at scale. We focus on challenges around security, efficiency, latency, and cost, and how we addressed these challenges in a system that combines caching, deduplication, convergent encryption, erasure coding, and block-level demand loading.

[AWS Lambda 10 Year Celebration Livestream](#)

Serverless Land (AWS YouTube channel) · Nov 2024

I represented the AWS Lambda engineering team on this livestream celebrating 10 years of the service, talking about how the service thinks about availability and making sure customer workloads of all shapes and sizes get a great experience. We also discussed how Lambda evolved into building and using Firecracker for virtualization and the challenges of doing a large scale migration.

PROJECTS

[Amazon Bedrock AgentCore Gateway tools](#)

AWS announcement for server-side tool execution through AgentCore Gateway integration with the Responses API.

[Amazon Bedrock server-side tool use](#)

AWS documentation for Responses API connectors using Lambda and MCP-compatible tools.

[Amazon Bedrock Web Search](#)

AWS announcement for built-in Web Search for OpenAI models in Amazon Bedrock.

[Amazon Bedrock server-side tools](#)

AWS announcement for server-side custom tools through the Responses API.

[Amazon Bedrock Responses API](#)

AWS announcement for OpenAI Responses API support on Amazon Bedrock endpoints.

[Project Mantle](#)

Amazon's public account of Mantle and the small engineering team that rebuilt the Bedrock inference engine.

[AWS Lambda on Graviton](#)

AWS reference for running Lambda functions on Arm-based Graviton2 processors.

[AWS Lambda SnapStart](#)

AWS announcement for Lambda SnapStart and its cold-start performance improvements.

[AWS Lambda container images](#)

AWS announcement for Lambda support for container images as a packaging format.

[Firecracker](#)

Official Firecracker project reference for lightweight virtualization used by AWS Lambda and other services.

Goodreads Deals

Public announcement describing the personalized Goodreads Deals ebook campaign.

Kindle X-Ray

Public announcement describing X-Ray for exploring characters, places, ideas, and topics across a book.

Goodreads on Kindle

Public announcement describing the original Goodreads integration on Kindle Paperwhite.

Kindle Fire launch

Public announcement describing the original Kindle Fire product and its launch.

Gawker

Open source application for creating parallel time-lapse movies using network-connected video cameras.

EDUCATION

University of Colorado

Five Year Program for Bachelors and Masters Degrees · Computer Science with a minor in Applied Mathematics

Jun 1998 – Jun 2003